25% STANDARD REDUCTION SHEET
For Offset Printing or Photostat Work
Finished Size 8" x 10-1/2"

Be sure that type on typewriter is clean. Use only black ribbon, or carbon ribbon if possible. All signatures, handwriting, or drawings must be done with BLACK ink.

[illegible] necessary to affix any copy to this sheet, [illegible] RUBBER CEMENT OR GUMMED BACK WHITE PAPER [illegible] paste or scotch tape.

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GGG-Y-35B
November 24, 1986
SUPERSEDING
GGG-Y-0035A(GSA-FSS)
August 13, 1964

FEDERAL SPECIFICATION

YARDSTICK (WOOD AND ALUMINUM)

This Specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers wood and aluminum yardsticks.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and classes. This specification covers the following type and classes when applicable, of yardsticks, as specified (see 6.2):

Type I. Wood (hard).
  Class 1. With flush or nonflush caps.
  Class 2. Without caps.

Type II. Aluminum.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications and Standards. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the General Services Administration, Specifications Unit (WFSIS), 7th and D Street SW, Washington, DC 20407.)

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Commercial Item Descriptions and the Index of Federal Specifications and Standards from established distribution points in their agencies.

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

FSC 5210

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3. REQUIREMENTS

3.1 Marking. Each yardstick shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined, part number, and country of origin.

3.2 Type I yardstick. Each type I yardstick shall be made of one-piece select, well seasoned straight-grained, hard wood. The yardstick shall be 36 inches long, within a tolerance of plus or minus 3/32 inch. The yardstick shall be 3/16 inch minimum in finished thickness and one inch minimum in finished width. Graduation lines shall be of uniform width and shall be filled with a durable black material. The graduations shall be of such depth as to be visible to the naked eye when the yardstick is observed edgewise; and the graduations and numbering shall be of such depth as to be felt when the inspector runs his fingers across the faces of the yardstick. Graduation lines shall be straight, permanent, and free from discontinuities and ragged edges. Graduation lines representing equal fractional aliquots of any scale, in descending order, shall be of equal height and readily distinguishable by the relative height from graduation lines representing other fractional aliquots. Figures shall be clear, permanent, free from ragged edges and breaks, and shall appear in a distinctive manner so as to be readily legible under all service conditions. One edge of one face shall be graduated in eighths of an inch and numbered consecutively in inches from 1 to 35. The opposite edge of the same face shall be graduated and numbered to indicate at least 1/4, 1/2, and 3/4 fractions of a yard. At the option of the manufacturer, additional fractions of a yard or additional scales may be placed on either side of the yardstick. The yardsticks shall read from left to right. The other face shall be the same except reading from opposite end or at the manufacturer's option: one edge - centimeters to 1/2 centimeters, other edge - not graduated. Figures shall be horizontal reading, that is, the figures shall be upright when the rule is horizontal. The scale error on any of the graduated edges shall not exceed plus or minus 1/32 inch between any two graduated dimensions or any dimension between either end of the yardstick and any graduation. The out of parallel errors of one edge to the other shall not exceed 1/64 inch in 36 inches. The out of square error on either end of the yardstick shall not exceed a 1/2-degree angle. The yardstick shall be finished with at least one natural clear durable coat of varnish or other equivalent protective coating which will not conceal the grain in the wood. At the option of the manufacturer, the yardstick shall be provided with either flush caps in accordance with 3.2.1, or nonflush caps in accordance with 3.2.2.

3.2.1 Class 1 (flush caps). The yardstick shall be fitted at both ends with a brass, steel plated or aluminum cap extending not less than 1/4 inch onto both faces of the yardstick. The yardstick ends shall be tenoned so that the cap faces are flush with the wood yardstick faces. Each cap shall be flush-riveted with at least two brass rivets or brass wire pins. Graduation lines are not required on the surfaces of the caps. The yardstick shall be provided with a 1/4 inch plus or minus 1/32 inch centrally drilled hole at each end for hanging purposes.

3.2.2 Class 1 (nonflush caps). The yardstick shall be fitted at both ends with a brass, steel plated or aluminum cap extending not less than 1/4 inch onto both faces of the yardstick. The yardstick ends shall be tenoned so that the cap faces are flush with the wood yardstick faces. A 1/4 inch plus or minus 1/32 inch inside diameter hollow brass rivet shall be used for fastening each cap; the rivet heads and cap rings may project above the flat surfaces of the yardstick. Each yardstick side shall be straight within 3/32 inch. Caps may be brass, plated steel or aluminum. Graduation lines are not required on the surfaces of the caps.

3.2.3 Class 2. Type I, class 2 yardstick shall be plain (without caps).

3.3 Type II. Each type II yardstick shall be made of one-piece properly tempered aluminum finished to minimize reflection, and shall have a 1/4-inch hole centrally located at one end for the purpose of hanging the yardstick. The yardstick shall be 36-inches long, within a tolerance of plus or minus 1/32 inch. The yardstick shall be at least 1/16-inch in thickness and at least one inch in width Graduation lines shall be straight and of uniform width. Graduation lines representing equal fractional aliquots of any scale, in descending order, shall be of equal height and readily distinguishable by the relative height from graduation lines representing other fractional aliquots. Figures shall be clear, permanent, free from ragged edges and breaks, and shall appear in a distinctive manner so as to be readily legible under all service conditions. Both edges (one face only or both faces) shall be graduated. One edge on one face shall be graduated to 16th of an inch and the other edge shall be graduated to either 8ths or 16ths of an inch, and the edges on the opposite face (when required) shall be graduated to the same requirements. Each yardstick shall be numbered consecutively from 1 to 35 (or 36) on the face(s). The figures (or numbers) shall be either horizontal-reading or vertical reading. Horizontal-reading yardsticks shall have two rows of numbers on the face(s). When the horizontal-reading figures are upright, they shall read from left to right. Vertical-reading yardsticks shall have at least one row of numbers on the face(s), with the number 1 starting on opposite ends of the yardstick. The scale error on any of the graduated edges shall not exceed plus or minus 1/32-inch between any two graduated dimensions, or any dimension between either end of the yardstick and any graduation. The out of parallel error of one edge to the other shall not exceed .016-inch in 36 inches. The out of square error on either end of the yardstick shall not exceed a 15-minute angle. Graduations and figures shall be printed black, with a durable ink. The yardstick shall be either anodized or coated with clear, abrasion resistant, baked, clear lacquer.

3.4 Workmanship. Workmanship shall be first class throughout. The yardstick shall be free from defects which may affect serviceability or durability. Burrs and sharp edges which might injure the operator shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in Military Standard MIL-STD-105, except where otherwise indicated herein.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements specified herein.

4.2.2 Inspection of the end item. Inspection of the yardstick shall be performed as specified in the following paragraphs. A lot shall consist of completely fabricated yardsticks of the same type, and class when applicable, offered for inspection at one time. The sample unit shall be one yardstick.

4.2.2.1 Visual examination. Each sample unit shall be visually examined for the defects set forth below. The inspection level shall be level I, with an acceptable quality level (AQL) of 2.5 defects per 100 units.

Examine	Defects
Finish	Not as specified
	Not clear finish, so as to conceal the grain in the wood (Type I only).
	Not smooth, clean, not free from burrs, sharp edges or other detrimental defects.
Material	Not fabricated from material specified.
	Cracks or splits.

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For Official [illegible]
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Examine	Defects
Construction and Workmanship	Ends not tenoned, so that cap faces are not flush or nonflush (as required) with wood yardstick faces. (Type I, class 1 only).
	Graduations or figures missing, illegible, not uniform, not readily distinguishable, not straight, ragged edges or not graduated as specified.
	Graduations and figures not impressed (Type I only).
	Different fractional aliquots not distinguishable by their relative height.
Marking	Missing, incorrect, not legible.

4.2.2.2 Dimensional examination. Each sample unit shall be examined for dimensions specified in applicable paragraphs. Any departure from specified requirements shall constitute a defect. The inspection level shall be level S-3 with an AQL of 1.5 defects per 100 units.

4.2.2.3 Testing. Each sample unit shall be tested as specified in paragraph 4.4.1. Any determination not within specified requirements shall constitute a defect. The inspection level shall be level S-4 with an AQL of 1.5 defects per 100 units.

4.3 Lot acceptance tests.

4.3.1 Scale accuracy, parallelism, and squareness of ends. The sample yardstick selected in accordance with 4.1.3 shall be subjected to the scale accuracy test of 4.4.1, to determine conformance with this specification. Any sample yardstick which does not conform to this specification shall be rejected, and if the number of nonconforming yardsticks in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Test procedures.

4.4.1 Scale accuracy. Graduated scales for linear measurement shall be tested for compliance with the accuracy requirements specified herein. These tests shall be made with suitable master rules, instruments, or laboratory apparatus of known accuracy. The error in any test measurement for accuracy shall not exceed the permissible accuracy error specified herein.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or C, as specified (see 6.2).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, and class when applicable (see 1.2.1).
(c) Levels of preservation, packaging, and packing required (see 5.1).

MILITARY INTERESTS:

NONE: DoD has determined that
no military activity has an official
interest in this Federal specification.

PREPARING ACTIVITY

GSA - FSS